

EXTREME HIGH POWER DENSITY T-MODULAR-MULTILEVEL-CONVERTER FOR MEDIUM VOLTAGE MOTOR DRIVE

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Abstract—The objective of the concept of T-type Modular Multilevel Converter (TMMC) is to improve the power density of the traditional Modular Multilevel Converter (MMC) exploited in medium voltage motor drive. The size of the major volume contributor “capacitor” can be reduced by 30%~40% with the proposed circuit operating at only half dc link voltage at low frequency low voltage region, which is enabled by an extra full bridge arm inserted between the middle point of the dc and the terminal of the output in each phase leg. It is also found to be a topology that is optimized for usage of SiC device because of its zero requirement for device serialization and its modest dv/dt and di/dt level

Keywords—TMMC, power density, SiC device

I. INTRODUCTION

The problem of the conventional VFDs for high power applications are bulky size. The proposed SiC-based TMMC is a high power density, high scalability, and Low-EMI modular multilevel converter, which is suitable for medium voltage drive. Compared to traditional 2-level (2L) converter and 3-level (3L) NPC, the bulky filter is removed. Compared to MMC, the power density is improved by at least 30%..

Take traditional 2L-converter, 3L-NPC, Cascaded-H-Bridge (CHB) and MMC for comparison. The first condition for comparison is with the usage of Si-IGBT. The problem of 2L and 3L converter is the bulky output filters because of the low switching frequency of the high voltage Si-IGBT they use. CHB and MMC can overcome this problem by increasing the number of modules and using lower voltage device in the modules. But the problem with CHB is that it requires a bulky low frequency multi-phase transformer and the problem with MMC is that it requires large volume of capacitor due to its fundamental current flow especially at low frequency

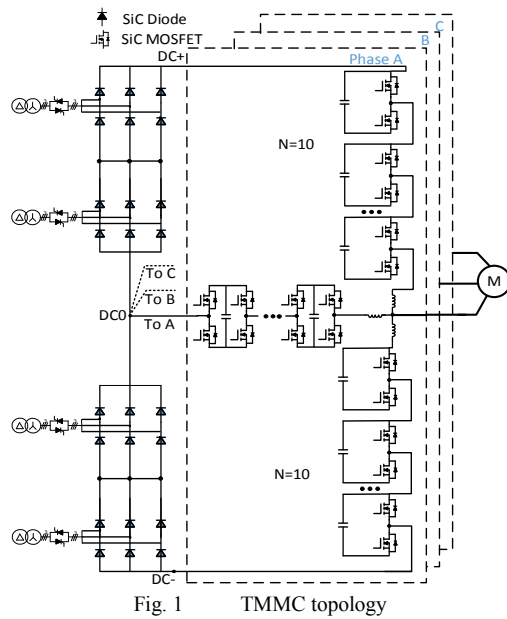
Then the question becomes whether the power density of the traditional 2L-converter, 3L-NPC, Cascaded-H-Bridge (CHB) and MMC can be improved by simply replacing the Si IGBT by SiC MOSFET rather than modifying the topology. So the second condition for comparison is with the usage of SiC-

MOSFET. For 2L and 3L converter, the filter size can be reduced by switching the SiC MOSFET at much higher frequency. However, it requires series connection of device to satisfy its voltage level since the maximum voltage rating of the current available SiC MOSFET is much lower than Si-IGBT. And the series connection can bring the following four problems: 1) the gate driving signal of the devices needs to be synchronized precisely; 2) the dv/dt level of series connected devices is the product of a single device dv/dt and the number of devices; 3) the circuit structure of the two-level and three-level converter requires the isolated gate driver for each device which brings high cross-talking noise; 4) the reliability of series devices is low. Therefore, if the SiC MOSFET is used in the conventional two-level or three-level converters, the dv/dt of a single device needs to be reduced by slowing down the switching speed of the device. This can be implemented by adding a big snubber circuit or extra miller capacitor. However, it sacrifices the benefits of fast switching speed feature of SiC device. The requirement for device synchronization also limits the maximum switching frequency of the device. The excessive voltage overshooting on the dc link will need additional redundant devices to be added, which lowers the device utilization rate, in another word, lowers the system efficiency. The proposed TMMC can address the above four problems since it is a series half-bridge/H-bridge module structure in which each module is rated according to the device rating, and no series device connection is needed. Thus the four problems becomes four advantages of SiC-based TMMC: 1) the module operates independently thus no device-level synchronization is required; 2) the dv/dt is at the single-device level, which is much lower than that of series devices, thus the dv/dt caused dc link voltage overshooting problem is mitigated to the minimum level 3) the cross-talking leakage current caused by the high dv/dt on two sides of the isolation transformer in the gate driver is significantly reduced because the gate driver power is drawn directly from the dc capacitor of the module and the highest isolation level is the isolation between the upper device and the lower device of a single half-bridge/H-bridge, which is equal to the dc voltage of a single module; 4) the system reliability is greatly improved and minimum redundancy number can be used to reach the reliability goal, because if one

module get shorted or broken, the system can continue to operate by bypassing the failure module. In summarize, the proposed TMMC topology is able to significantly reduce the capacitor size and is quite appropriate to be used with SiC device since it overcomes the major disadvantage of SiC MOSFET such as high EMI noise, high common mode current, and low reliability. In addition, the use of high voltage SiC-MOSFET can also lower the total module numbers to conquer the disadvantage of TMMC. In a word, the concept of TMMC is to use more active devices to reduce or substitute the passive components, mainly the bulky capacitor

II. PRINCIPLES

A. Topology and principles

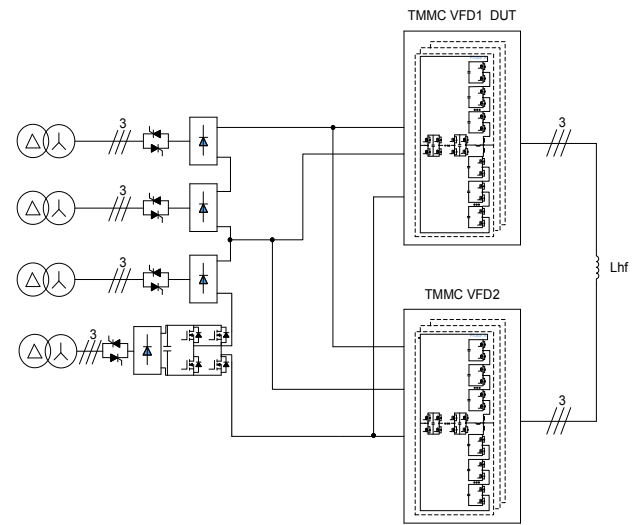


The proposed TMMC can be applied in medium voltage drive. In this application, the output is in a constant V/f control, and the torque is a constant at the whole operating range.

The first innovativeness of the proposed topology is that it creates a middle arm, which is composed of cascaded H-bridge and connects to the middle point of the front-end rectifier, to make the converter operate under half dc link when the output voltage is below 0.5 p.u. When the output voltage is above 0.5 p.u., the upper and lower arm is in usage and the operation is exactly the same as MMC. When the output voltage is below 0.5 p.u., one of the upper arm and lower arm is used with the middle arm. In this case, only half dc voltage is supplying the power. So the arm voltage is reduced by half because the arm voltage is equal to half dc voltage minus the ac voltage. The arm current is equal to one third of the dc current minus half output current, and it increases by a very small percentage because the output current remains constant as rated. The dc current increased to two times but this increase can be ignored because the dc current is much smaller than the ac current at low power low frequency operating point. Therefore, the fundamental frequency energy ripple in one arm is reduced to

half of the original. The capacitor size which depends on the energy ripple amplitude is also reduced by half. Because of the capacitor installation in the extra arm, the overall capacitor size is reduced by 30%. The system power density is improved by 30% because the capacitor size takes up 90% of the total volume.

The second innovativeness is that this T-type MMC does not use any extra thyristor arm or series IGBT arm to create this half dc link voltage and the extra arm still has the same modular structure as the original arm. The only difference is that the half bridge is replaced by H-bridge because the middle arm needs to undertake the voltage in both directions. So all the advantages of modular structure such as low dv/dt, low common mode noise, low EMI, high scalability, low THD, high reliability can be kept.



The third innovativeness is that this TMMC can start the motor from very low frequency easily by generating a ramping dc voltage from zero, when bypassing three of the diode bridges and using only the fourth diode-bridge and H-bridge, as shown in Fig. 2. At a wide frequency range of [30Hz, 150Hz], the dc link voltage keeps a constant. Only at the start-up, when the frequency is below 30Hz, the voltage is lowered down by bypassing two diode bridges with turning off the corresponding thyristor pair. At extremely low frequency of [0, 15Hz], the dc voltage is ramping up linearly from zero by using only the fourth diode bridge and its corresponding H-bridge. The output ac voltage and frequency, the input dc voltage are shown in Fig. 3 (a). Because of the use of the middle arm, the actual utilized voltage by the inverter is reduced to half dc voltage when the frequency is lower than 30Hz, which is shown in Fig.3 (b). By using this starting-up strategy, the maximum capacitance requirements can be always smaller than the capacitance calculated at 0.2 p.u. frequency. The reason why not directly use diode-bridge to generate the half dc link or $\frac{3}{4}$ dc link in the high frequency range is that the slow response of thyristor cannot satisfy the dynamic requirement of the load in the normal frequency range. The usage of middle arm is a usage of semiconductor devices to

response to the dynamic frequency change, so it has enough speed to maintain the normal operation. The reason why not using one H-bridge with each diode bridge to create a continuous ramping dc link (the method Siemens Uses) is that the extra H-bridges also requires a large number of devices (mostly 4.5kV IGBT if the dc bus is 10kV), and large volume of capacitors (single phase H-bridge). The proposed TMMC utilize the two original arms in the inverter and keeps the dc link a constant value during a wide frequency range. It has much simpler modular structure, faster dynamics and simpler control architecture.

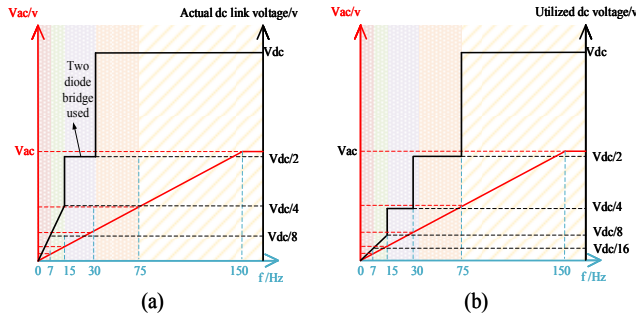


Fig. 3 Motor operation V/f curve and the corresponding actual dc voltage (a) and utilized dc voltage (b)

B. Performance matrix

The full-scaled medium voltage drive system is rated at 10kV DC, 6.6kV ac and 2MW. The final delivery for the project is a down-scaled pump-back system which is composed of two 600kW, 3kV DC to 1.5kV ac, 0~150Hz TMMC inverters with 1.7kV/225A SiC MOSFET (Wolf speed CAS300M17BM module or three parallel 1.7kV/72A TO-247 MOSFETS (C2M0045170D)). This power proportionally down-scaled prototype can accurately predict the power density and efficiency of the actual 2MW/10kV system. The power density of the two are similar as well because the capacitor size is proportional to power rating and it takes up more than 80% of the total volume.

The technical goals of the motor drive converter is to achieve 150W/inch³ power density, ≥ 97.5 efficiency at full rating and $\geq 95\%$ at 5% rating, $\geq 2x$ over SOA MTTF, and $\leq \$0.5/W$ cost. MMC is considered as a strong competitive object to be compared with the target and TMMC here. The data from Siemens developed SINAMICS SM120 CM series medium voltage drive has been used in table I. The product series is a 3.3~7.2kV/ 6000~13700kVA 0~150 Hz MMC motor drive converter, with the efficiency of 98.5%. The table I shows the quantitative metrics that compares the proposed TMMC to the traditional MMC and the technical performance targets, at the 2MW full scale system. It indicates that TMMC has 40%

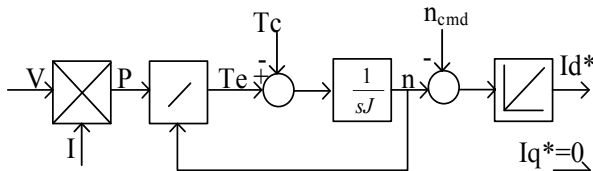


Fig. 4 Control strategy to simulate the motor operation of converter

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improvement on power density than MMC

TABLE.I Quantitative metrics for the traditional MMC and the proposed TMMC

Cat.	Metric	MMC	TMMC
Application	Medium Voltage Drive		
	Power (W)	2MW	
	Voltage (V):	10-kV dc and 6.6kV ac	
	Efficiency (%) and Q	>99% at rated power; =96% at 5% rated power	
	Redundancy	N+1	N+1
	Power density (W/inch ³)	169	255
	Relative cost (\$/W)	0.022	0.021
	Approximate MTTF	2.5 million hours	1.6 million hours
	Cooling methodology	water cooling	
	EMI filter volume	3%	3%
Circuit level	Num. of discrete modules	66	102
	Switching methodology	interleaved PWM	
	Switching frequency (kHz):	20	
Component level	Num. of discrete drivers	66	102
	Transistor type(s) & count	CAS300M17BM2 Count: 66	CAS300M17BM2 Count: 102
	Transistor specification	1.7-kV/225A half-bridge	
	Total Semiconductor area (mm ²)	433752	670344
	Total inductor volume	1% of total volume	
	Capacitance/module	896 μ F	426 μ F
	Capacitor type and count	1300V/28A/40 μ F B32778G1406 Quantity: 23	1300V/28A/40 μ F B32778G1406 Quantity: 11
	Total cap. volume (mm ³)	224*10 ⁶	165*10 ⁶

III. SYSTEM DESIGN

A. Pump-back system design

A pump-back system will be built to simulate the motor drive operation. Converter 1 is controlled as the motor drive, and converter 2 is controlled to simulate the motor operation, as shown in Fig.4. An inductor is inserted between the inverters at the ac side. They share the dc link, which is provided by four series diode-bridges.

The one H-bridge is used to ramping up the dc voltage linearly from zero to $\frac{1}{4} V_{dc}$. Converter 1 will be controlled to output rated real power and converter 2 will absorb all the real power. The dc link current of the two are also opposite. Thus the grid only provides the power loss.

TABLE.II SPECIFICATION OF THE DOWN-SCALED SYSTEM AND FULL-SCALED SYSTEM

Specification	Down-scaled system	Full-scaled system
Device V rating (V)	1700	1700
Device I rating (A)	225	225
Modulation index	1	1
power factor	1	1
modules#/arm	3	10
Power rating (kW)	600	2000
V _{dc} rating (V)	3000	10000
I _{dc} rating (A)	200	200
V _{ap} rating (V)	1500	5000
I _{ap} rating (A)	267	267
ω rating (rad/s)	942	942
Module voltage (V)	1000	1000
arm current rating (A)	200	200
Energy ripple peak/Arm (J)	511	1703
Energy ripple/module (J)	170	170
Module Capacitance (μ F)	426	426

The experiment is carried out under the down-scaled rating. The specifications of the down-scaled system and the full-scaled system are compared in table II. It shows that the down-scaled system has exactly the same voltage and current rating, the same energy ripple, the same capacitance as the full-scaled system. So the volume of a single module is the same as full-scaled system and its total volume is proportional to the number of modules, in another word, proportional to the power rating. Therefore, it can be used to estimate the power density of the full-scaled system. In terms of efficiency, the total power loss of the down-scaled system is also proportional to the number of modules, in another word, proportional to the power. Therefore, it can also be used to estimate the efficiency of the full-scaled system.

B. Half bridge module design

1): Device rating, number of modules and module voltage level selections

1.7kV SiC MOSFET and 1.2kV SiC MOSFET are compared in terms of system efficiency and volume. The results are shown in Fig. 5 and Fig.6.

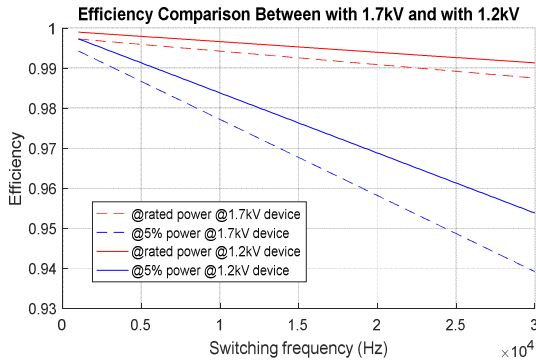


Fig.5 Efficiency vs. Switching frequency for 1.7kV and 1.2kV

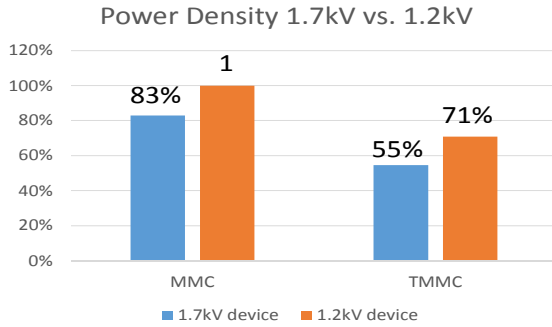


Fig.6 Power density comparison between 1.7kV and 1.2kV device

In terms of efficiency, the voltage utilization of device affects the conduction loss and the device switching characteristics affects the switching loss. The switching loss dominates the loss in the case of low number of modules. The results show that 1.2kV device has much lower switching loss than the 1.7kV device. So the system with 1.2kV device has higher efficiency. In terms of volume, the higher the device rating, the less series capacitor is needed and each capacitance becomes smaller. So the system with 1.7kV device has higher

power density. Since the efficiency of the system with 1.7kV device already exceeds the standard, the volume becomes the main target. Therefore, 1.7kV device is selected.

2): DC capacitor calculation and selection

The DC capacitance in a single module is calculated according to the energy ripple. The energy ripple in one module is equal to the energy ripple in one arm divided by the number of modules. The energy ripple in one arm is equal to the integration of arm voltage and current product over one fundamental period:

$$E_{arm} = \int_0^{T_s} V_{arm} \times I_{arm} dt \quad (1)$$

The energy ripple will be revealed on the capacitor voltage ripple, so a certain capacitance is needed to keep the voltage ripple within a limit. Assume the allowed maximum voltage ripple is ΔV and the nominal capacitor voltage is V_{norm} . The minimum capacitance satisfies the following relationship:

$$\frac{1}{2} \times C_{min} \times ((V_{norm} + \Delta V)^2) = \frac{E_{arm}}{N} \quad (2)$$

in which C_{min} stands for minimum capacitance, and N stands for the number of modules per arm.

The voltage rating of the capacitor needs to be a little higher than $(V_{norm} + \Delta V)$. The current in the capacitor is equal to the arm current, and its rating should be 1.2~1.5 times of the current. An operating point needs to be selected for capacitance calculation. According to the energy equation, it can be seen that the maximum energy ripple occurs at near-zero frequency. But the variable dc link will be used to start the motor, and the operating point to calculate the capacitance is shown in Table III.

TABLE.III OPERATION POINT TO CALCULATE THE CAPACITANCE

	Power	Frequency	Vpeak	Ipeak	Vdc	Idc
P.U.	0.2	0.2	0.2	1	1	0.2
Absolute value	120kW	30Hz	300V	267A	3kV	40A

3): Switching frequency selection

The switching frequency is designed according to the following items: (1) target efficiency at full power and 5% power; (2) minimum pulse calculated from and of the single device (3) control system communication loop time (4) allowed cooling size and weight (5) output THD requirement. After checking all these factors, it is found that the efficiency is the bottle-neck of the frequency. The loss of the device dominates the total power loss and it can be calculated as:

$$P_{switching} = \frac{(E_{on} + E_{off})}{V_{test} \times I_{test}} \times V_{opr} \times I_{opr} \times f_{sw} \quad (3)$$

$$P_{cond} = I_{rms}^2 \times R_{dson} \quad (4)$$

In order to achieve 95% efficiency at 5% rated power, the frequency is limited to 20kHz, and the efficiency for the full rating is 99% at this frequency

IV. CONTROL ALGORITHM AND SIMULATION RESULTS

A. Starting-up strategy

The starting-up strategy is shown in Fig. 3. Fig. 3 (a) shows the dc voltage generated from the diode bridge (black) and the output voltage which is proportional to the frequency (red). Fig. 3 (b) shows the actual utilized dc voltage due to the new topology (black) and the output V/f curve. In the full frequency range, the output current is always equal to the full rating. The output voltage is proportional to the frequency and the dc voltage keeps constant in the frequency range of [0.2 p.u., 1 p.u.], but the utilized voltage at [0.2 p.u., 0.5 p.u.] is only half dc link because only the upper arm and the middle arm is operating. For the frequency range of [0.1 p.u., 0.2 p.u.], the dc voltage is stepped down to $\frac{1}{2}$ Vdc by bypassing two diode bridges in the front end, but the utilized voltage is only $\frac{1}{4}$ Vdc. For the frequency range of [0 p.u., 0.1 p.u.], the dc voltage is ramping up linearly from 0 to $\frac{1}{4}$ Vdc by using the fourth diode bridge and H-bridge together and bypassing the other three diode-bridges, but the actual utilized voltage is from 0 to $\frac{1}{8}$ Vdc. It is obvious that the dc voltage has a much higher utilization rate with the new topology, which reduces the arm energy ripple.

The capacitance needed for each operating point is shown in Fig. 5. From this figure it can be seen that the capacitance designed for 30Hz (0.2 p.u.), half DC link voltage is able to satisfy the ripple requirement during the whole operating

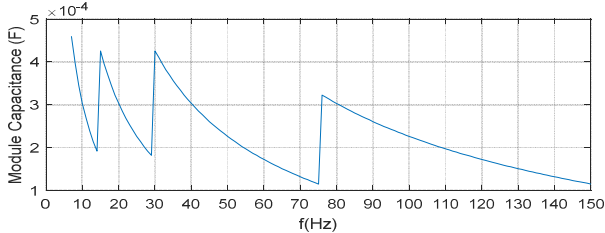


Fig.7 Required capacitance vs. frequency

range.

This voltage step-up is implemented by the series diode bridge circuit shown in Fig. 2. Each diode bridge can be controlled to output dc voltage by turning on the anti-parallel thyristors, or be controlled to output zero voltage by turning off the thyristors. The linear dc voltage from 0 to $\frac{1}{4}$ Vdc is generated by the H-bridge in the bottom diode-bridge.

B. Control strategy in normal operation and whole simulation results from start-up

1) Level 1: Pump-back system control

The final delivery will be the pump-back system shown in Fig. 2. The DFE only needs to provide the loss to the pump-back system. So the first level control is to implement the diagram shown in Fig. 4. The pump-back system simulation with TMMC has already been conducted and linear V/f with a

constant torque is implemented at the ac side. The results are shown in Fig. 8.

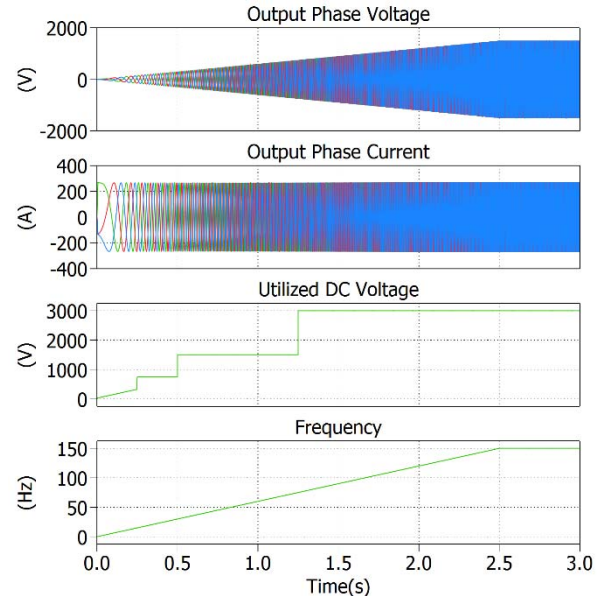
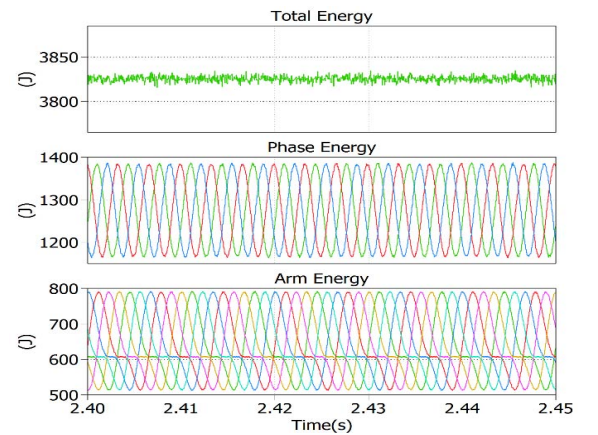


Fig.8 Simulation results of the pump-back system with the start-up

2) Level 2: Energy control loop

TMMC structure, same as MMC is an inverter with energy storage. The dc link capacitor voltage is a non-linear control plant. The outer control loop should be based on energy. Total energy balance, phase energy balance and arm energy balance are three control objects.

The total energy balance can be controlled through controlling the dc current when ac current is commanded by the power, or through controlling the ac current when dc current is commanded by the power. The phase energy balance can be controlled through the common mode circulating current which flows among phases. The arm energy balance can be controlled through the differential mode circulating current which flows between arms but not among phases. The definition of these two types of circulating current are described in subtask 3 of this task. And the initial simulation results about the three energy balance loops are shown in Fig. 9.



The diagram of the energy loops is shown in Fig. 10

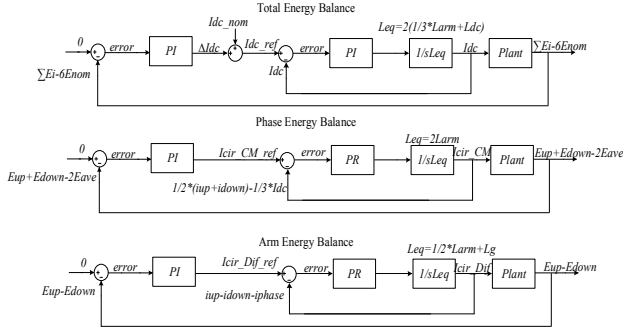


Fig.10 Energy control loops

3) Level 3: AC current, DC current, circulating current control

Since the system is operating with only six arms at any moment, the equivalent circuit model can be drawn with only six arms as shown in the left figure of Fig. 11. The arm current consists of four parts, common mode dc current, differential mode ac current, common mode circulating current among phases and differential mode circulating current between arms. These four currents are driven by four components in the arm voltage respectively: common mode dc voltage, differential mode ac voltage, common mode circulating voltage and differential mode circulating voltage. The equivalent circuit for each corresponding pair of voltage and current is shown in the right figures of Fig. 11. The simulation results are shown in Fig. 12.

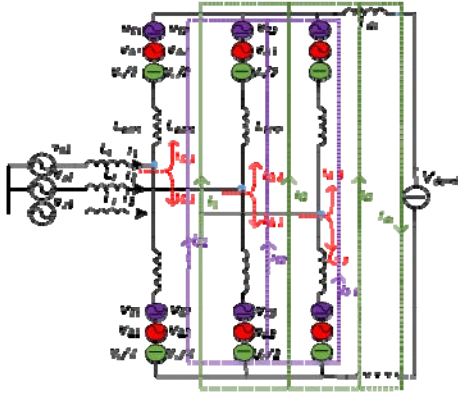


Fig.11 Equivalent circuit model of TMMC at different operating region and four voltage components in arm voltage

The next step development will be the design of the control parameters and the deep simulation including every details. In the final simulation, all the non-idea situation such as delay, line impedance, the real device model will be considered.

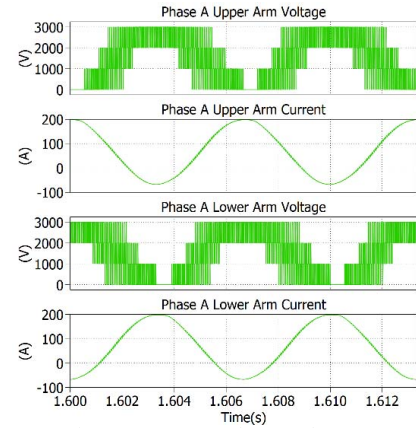


Fig.12 Arm voltage and current

4) Level 4: Capacitor balancing control

For the capacitor balancing, since the number of modules is small, “sorting” method is used. In this method, all the module voltage in one arm will be sorted and the highest voltage module is always be selected to be discharged and the lowest voltage module is always selected to be charged. The switching mode of the circuit is built in the simulation and the balancing results are shown in Fig. 13.

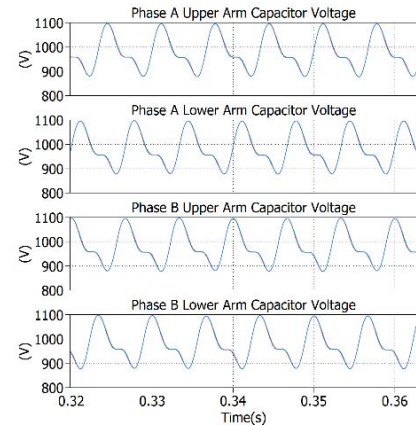


Fig.13 Capacitor voltages

V. CONCLUSION

The proposed TMMC a high power density, high scalability, low EMI, low THD and low common mode noise modular multi-level converter medium voltage motor drive. It improves the power density of traditional MMC drive by 30% while maintaining the high efficiency, low THD and the capability of starting the motor from zero frequency.

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